

A Multi-Parametric Deep Learning Approach for Short-Term Load Forecasting and Grid Anomaly Characterisation in Smart Grids

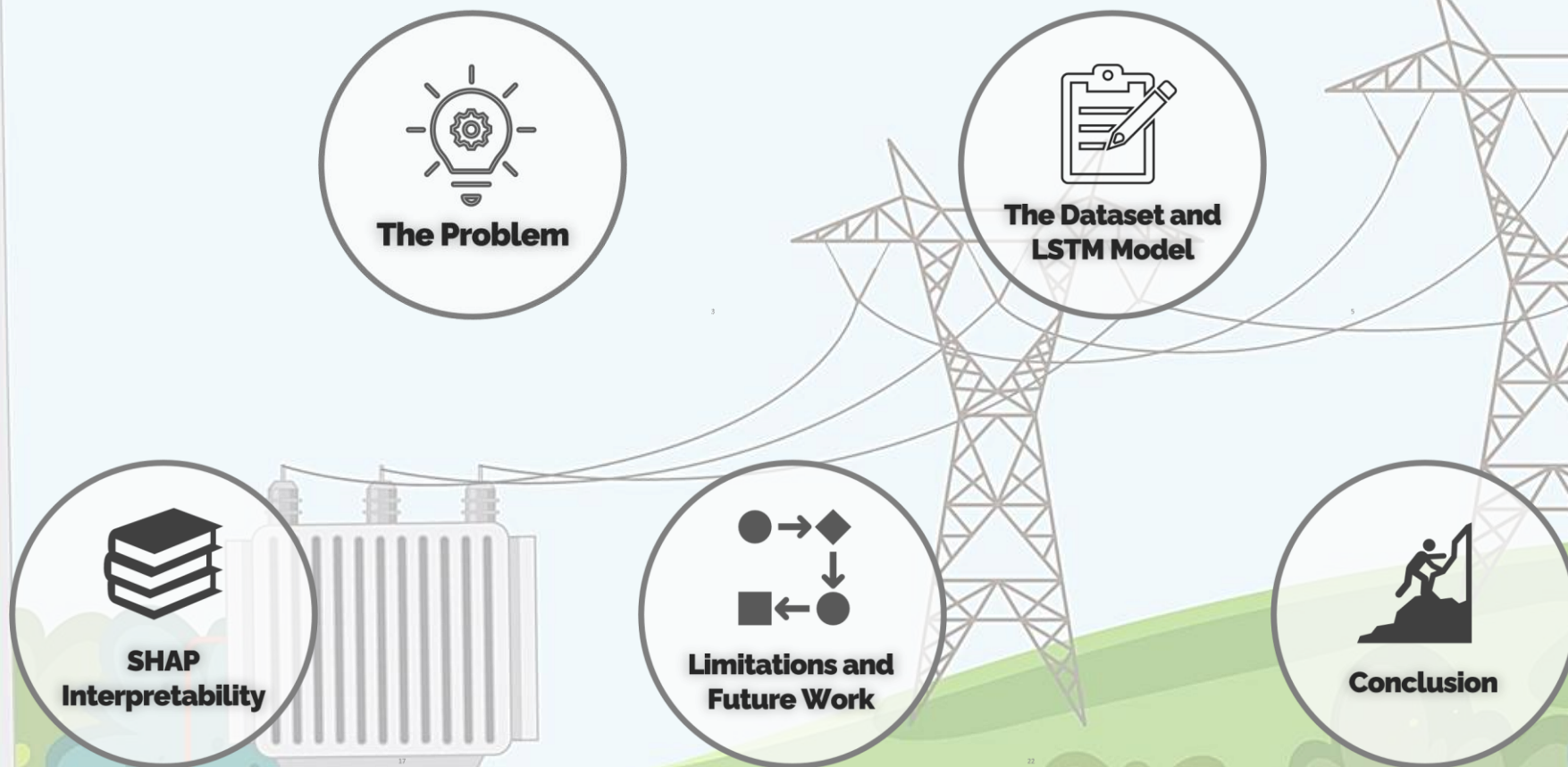
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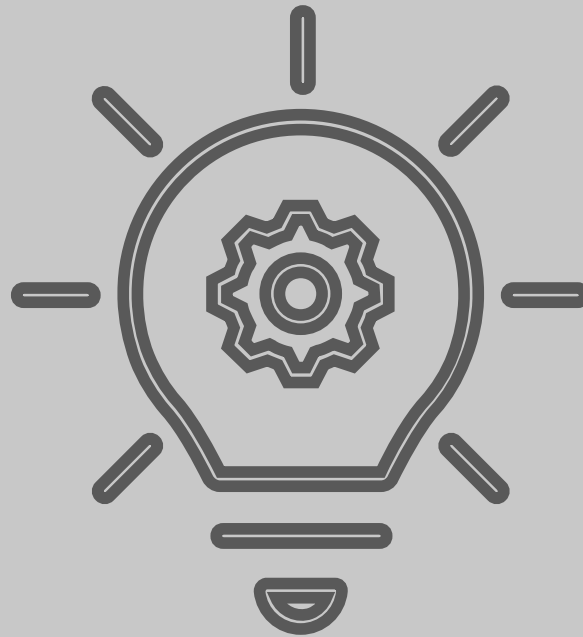
Saad Abdullah Majad

Student ID:

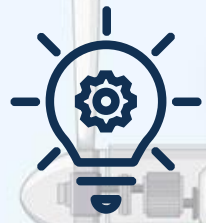
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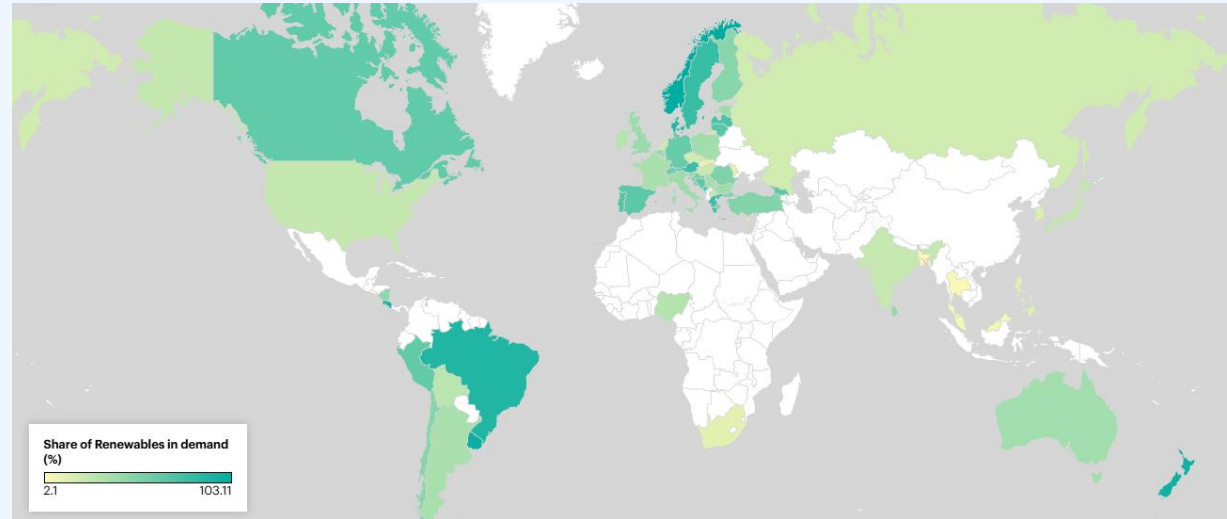
The Problem



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- Accurate load forecasting is essential for safe, efficient grid operation
- Existing literature is dominated by North American and European grids
- Transformer-level, sub-hourly, multi-parametric data from South Asia is rare



The Objective

- Build and evaluate a multi-parametric LSTM for transformer-level short-term active power forecasting
- Apply SHAP to make the model's reasoning transparent



The Dataset and LSTM Model



The Dataset and LSTM Model

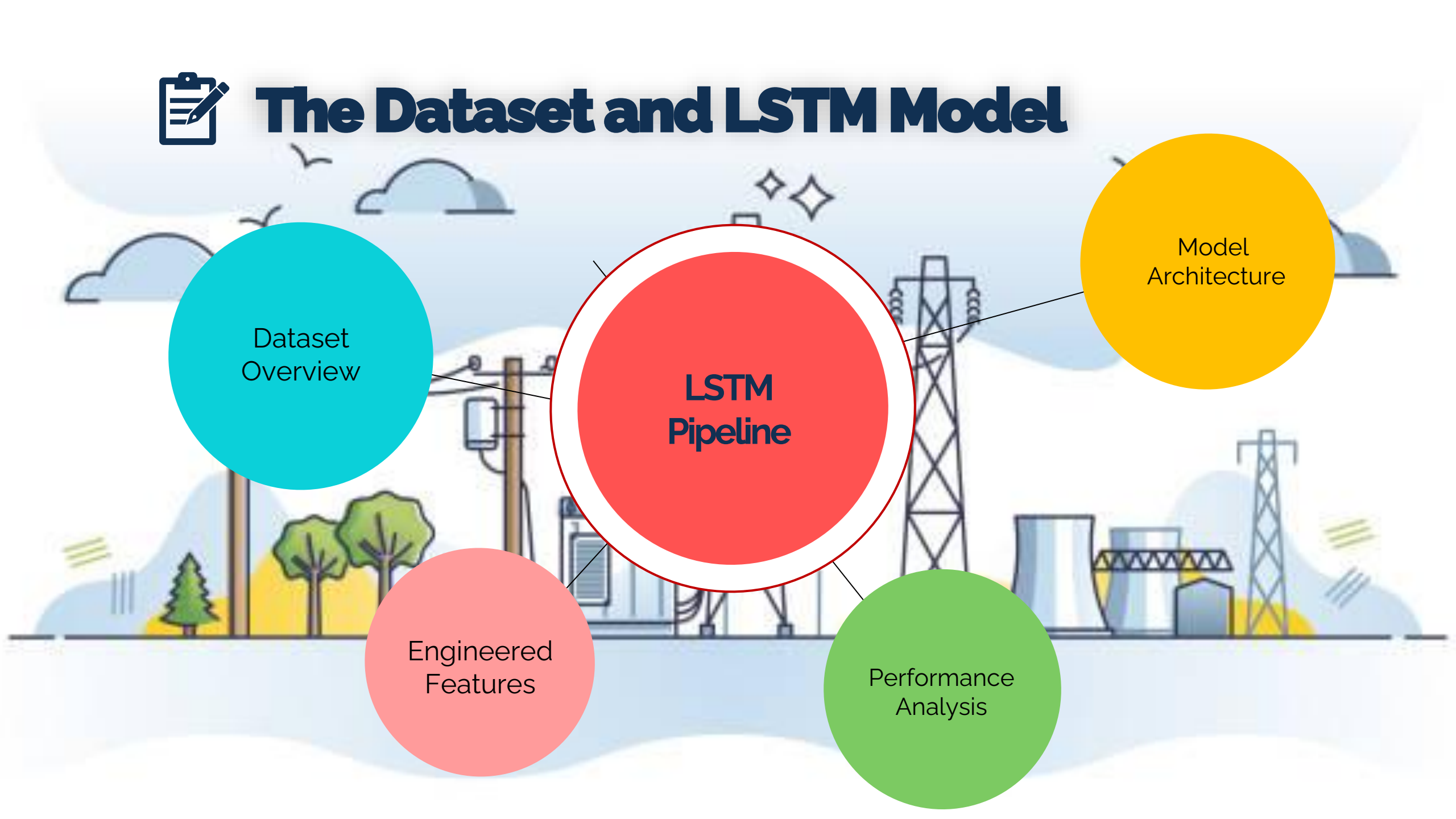
Dataset
Overview

LSTM
Pipeline

Model
Architecture

Engineered
Features

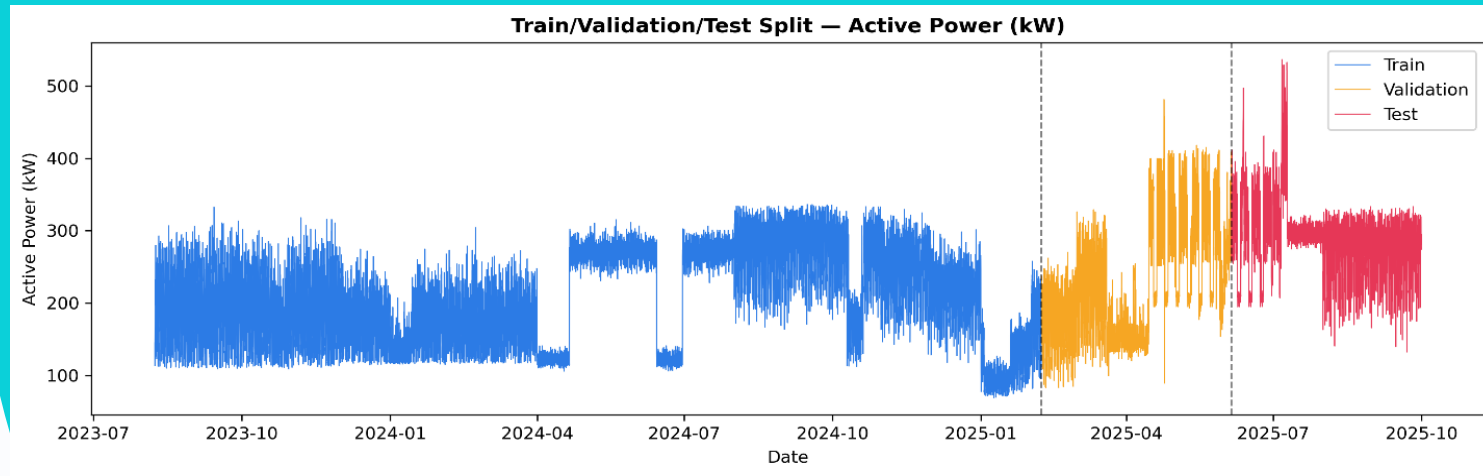
Performance
Analysis



The Dataset

The data is collected from a substation of the Gopalganj University of Science and Technology, Bangladesh.

- August 2023 – September 2025
- Three-phase currents, voltage, power factor + co-located weather data
- 19,008 hourly observations
- Chronological 70/15/15 train/validation/test split



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RangeIndex: 19008 entries, 0 to 19007
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Month                 26 non-null    object
1   Date                  19008 non-null object
2   Time                  19008 non-null object
3   System_V              19008 non-null int64
4   Current_R_A           19008 non-null int64
5   Current_Y_A           19008 non-null int64
6   Current_B_A           19008 non-null int64
7   pf                    19008 non-null float64
8   Notes                 3120 non-null  object
9   Temperature_C         19008 non-null float64
10  Humidity_%            19008 non-null int64
11  Precipitation_mm      19008 non-null float64
12  Wind_Speed_mps        19008 non-null float64
13  Solar_Radiation_Wm2    19008 non-null int64
dtypes: float64(4), int64(6), object(4)
```

Figure: Split of data for the LSTM model and dataset parameters

Insights with Anomaly Characterisation

These features were engineered for exploratory data characterisation to analyse physical grid realities.

Category	Characterised Event	Size	Implication
Outlier Processing	Phase Current Clipping	43 total points clipped	Preserved unbroken time-series continuity for autoregressive lags
Phase Imbalance	Phase Divergence	5,006 timestamps	Characterised structural phase asymmetry across transformer legs
Load Anomalies	Overload Peaks/Off-Peak Low Load	27 overloads/ 553 low load	Expected outages and non-linear peak load shocks
Voltage Dynamics	Voltage Sags	238 events	Identified severe voltage drops using empirical deviation thresholding



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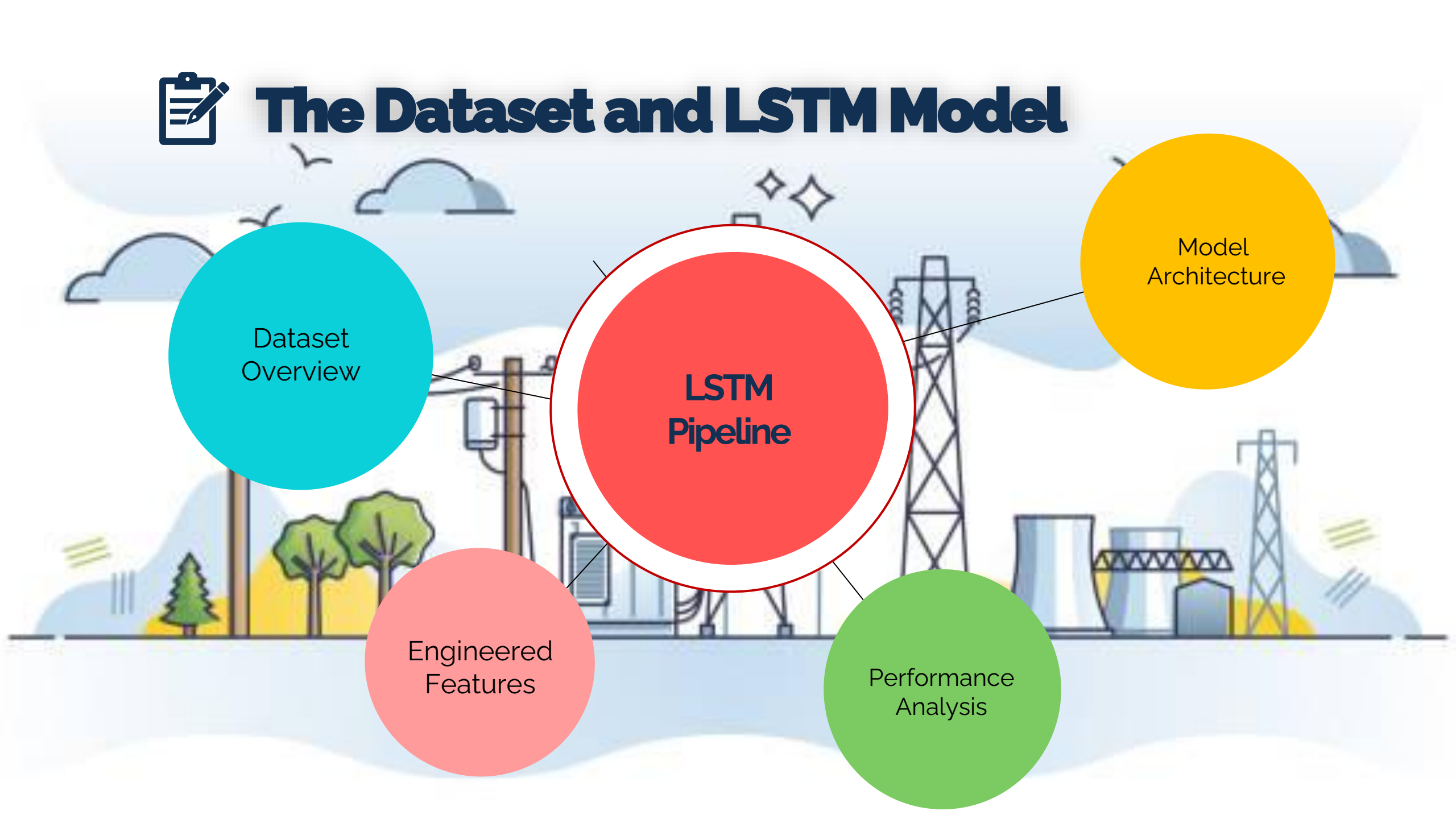
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Engineered Features and Selection

Active Power was calculated and set as the primary forecasting target, while candidate features were iteratively engineered, evaluated, and dropped to construct the final dataset.

$$P_{\text{active}}(\text{kW}) = \frac{\sqrt{3} \times V_{\text{system}} \times I_{\text{mean}} \times \text{pf}}{1000}$$

$$\text{CDH} = \max(0, T_{\text{ambient}} - 24.0)$$

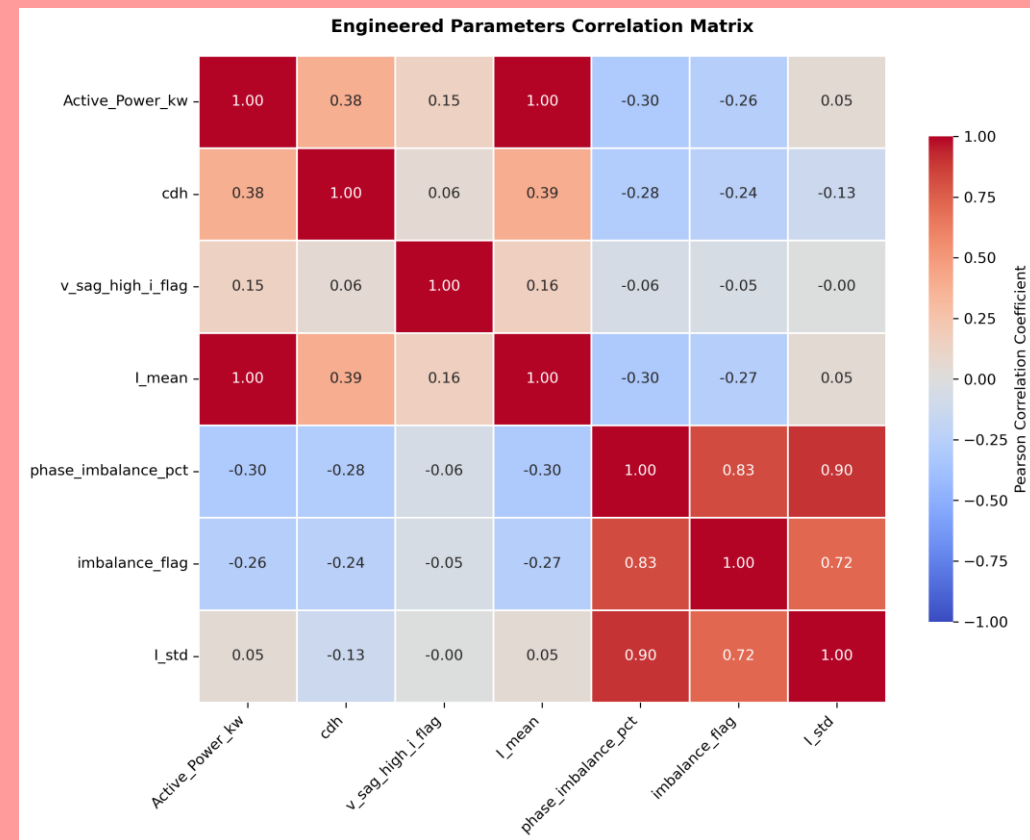


Figure: Engineered parameters correlation matrix



The Dataset and LSTM Model

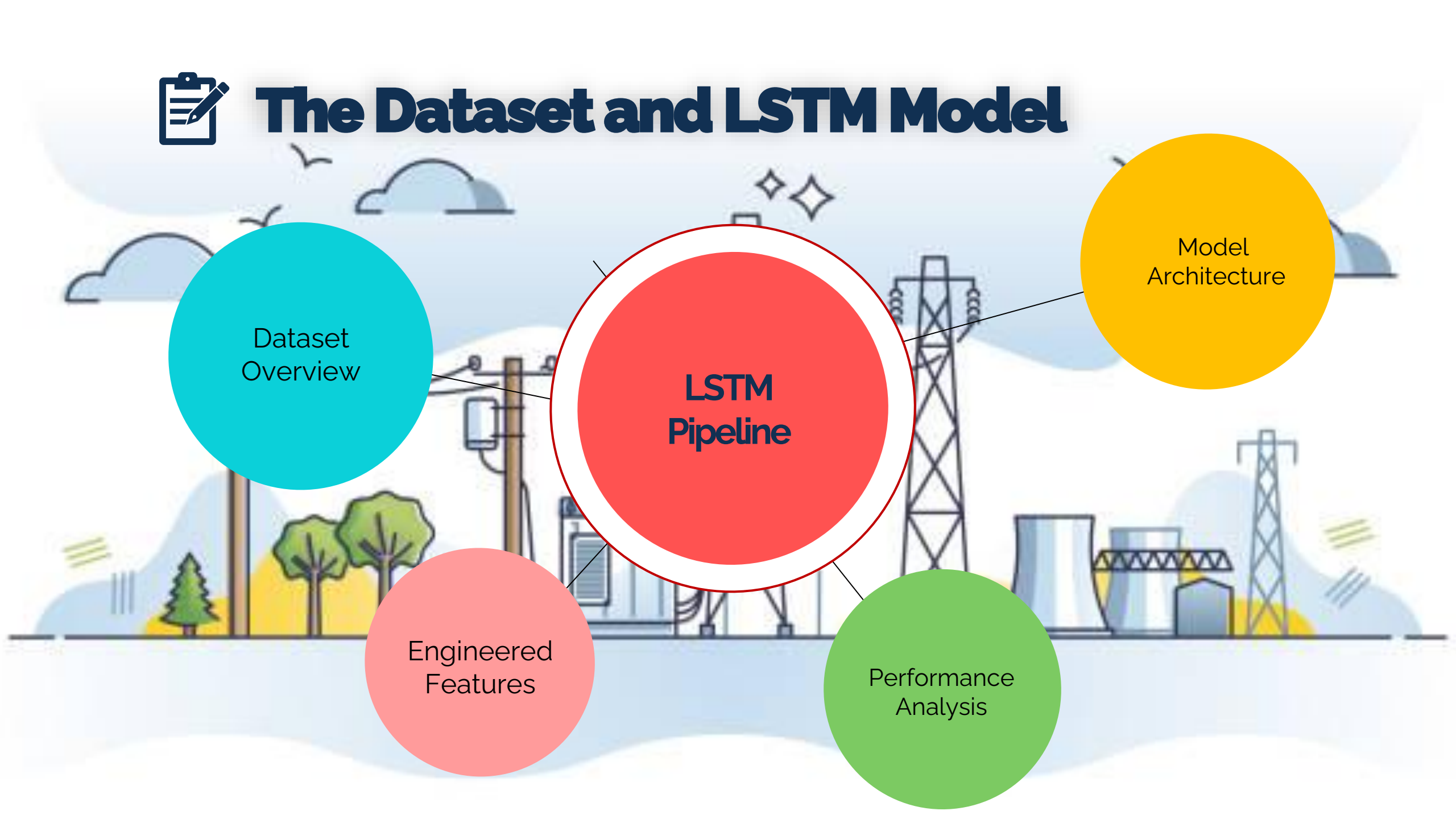
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Model Architecture

Data Pre-processing



Architecture

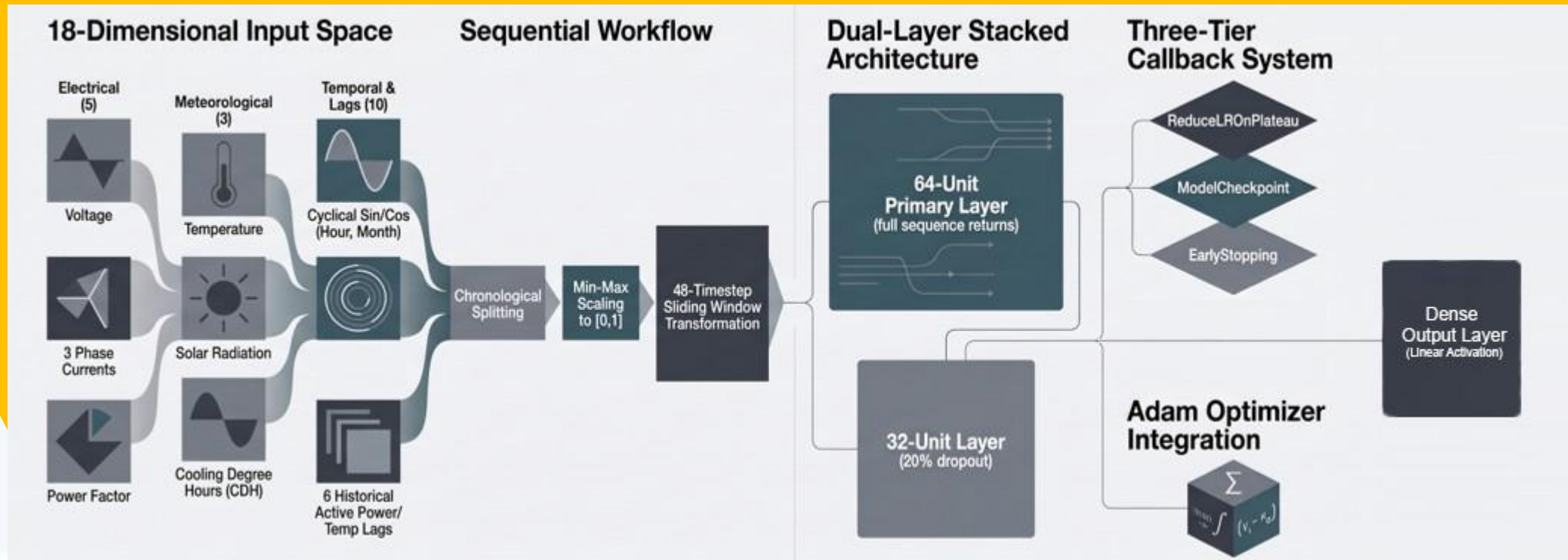


Figure: LSTM Model Architecture

Architecture Comparison

Configuration Number	Configuration	MAPE	Persistence MAPE
1	arch_64_32_dr0.2 (final model)	8.87%	8.83%
2	lookback_48	10.24%	8.83%
3	lr_0.0001	10.42%	8.83%
4	lookback_24	10.98%	8.81%
5	arch_128_64_dr0.3	11.19%	8.83%
6	lookback_72	11.30%	8.84%
7	lookback_48_plus_rollstd24	11.59%	8.83%
8	arch_128_64_dr0.1	11.60%	8.83%
9	arch_256_128_dr0.2	12.08%	8.83%
10	batch_32	12.71%	8.83%
11	lr_0.001	12.81%	8.83%



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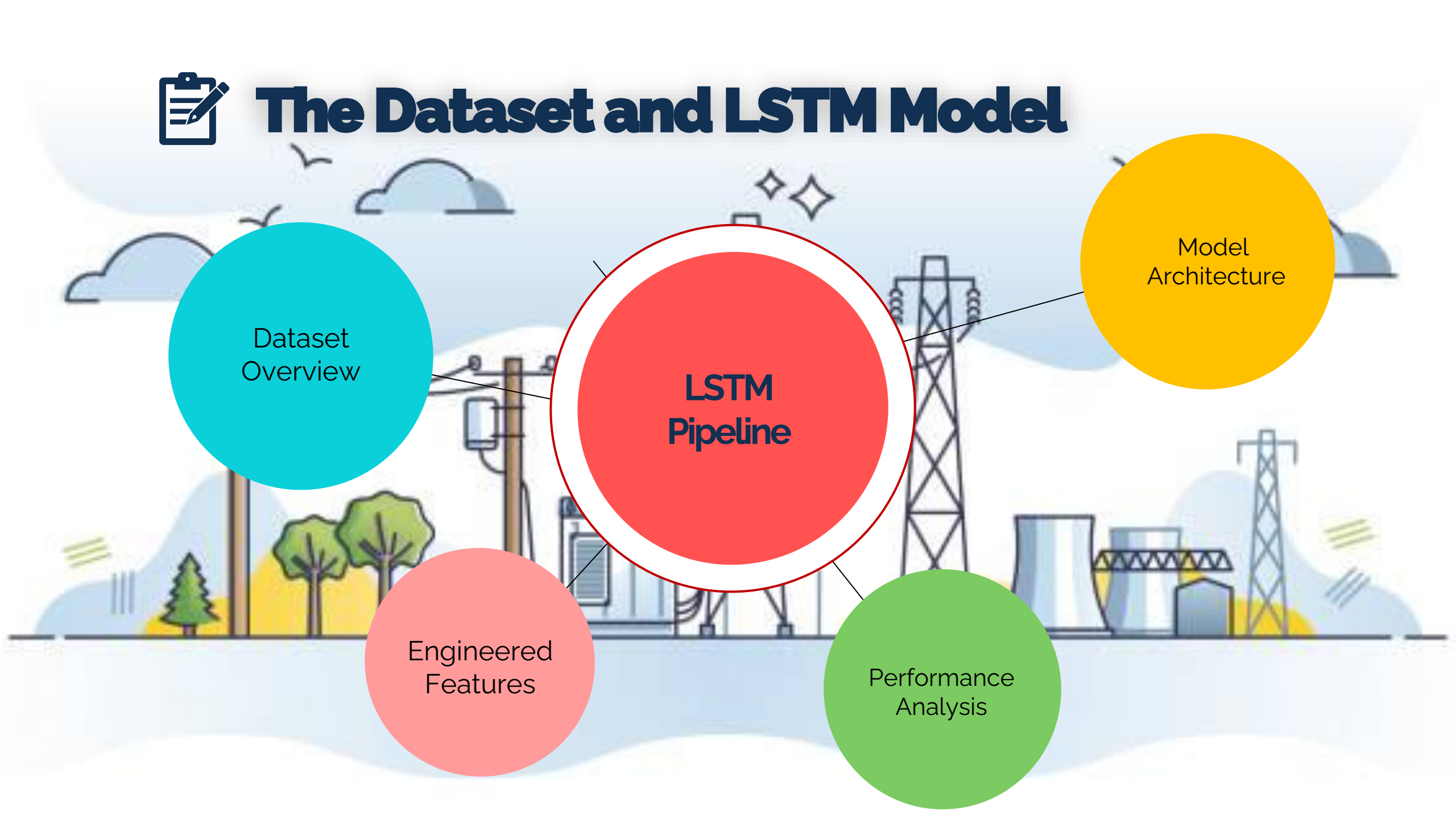
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Headline Results

Overall Forecast

Model	RMSE (kW)	MAE (kW)	MAPE (%)
Persistence (Naive Baseline)	33.68	24.18	8.83%
Stacked LSTM (Proposed)	35.23	25.13	8.87%

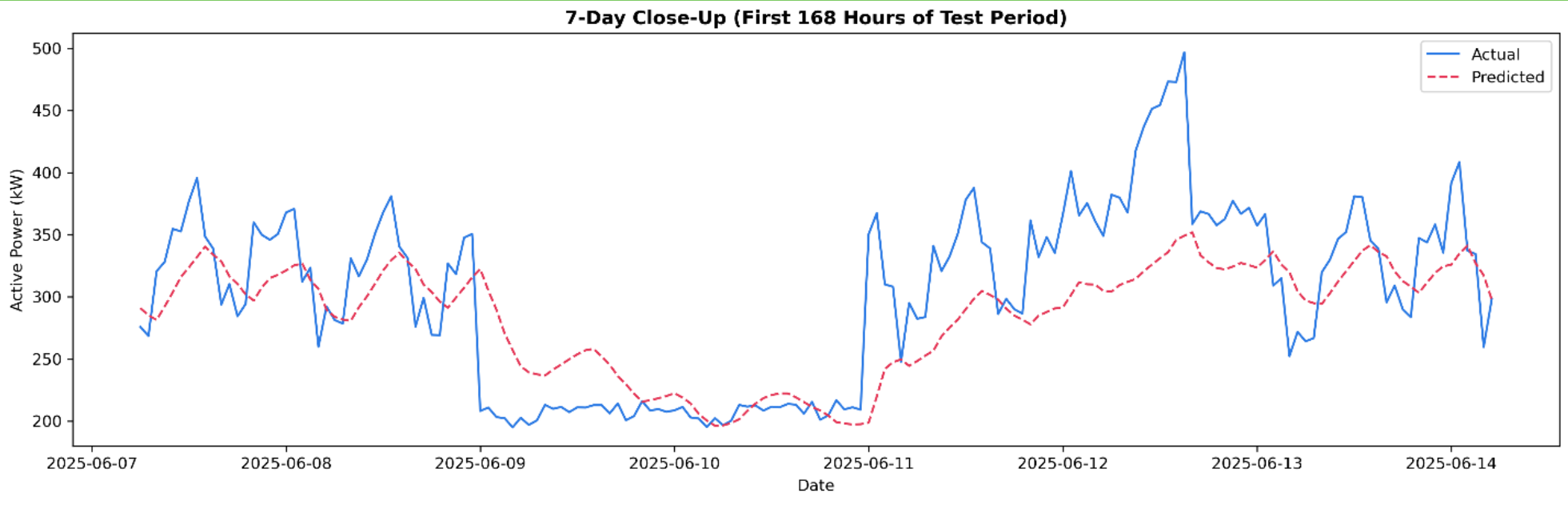


Table: Comparative Analysis for Proposed Model and Naïve Model
Figure: 7-day Performance of LSTM Model with Actual Power Use

Regime Dependent Results

Strengths

- During 68% of the test period (Calm window) the LSTM outperforms persistence

Weaknesses

- Multi-hour lookback and dropout regularisation smooth its predictions
- Brief lag around sudden spikes

Operational Segment	n	LSTM MAPE	Persistence MAPE	Dominant Model
Calm Window ($\geq 2025-07-14$)	1,896	7.65%	9.66%	LSTM (+2.01%)
Volatile Window ($< 2025-07-14$)	882	11.50%	7.04%	Persistence (+4.46%)
Term-Time	2,568	8.65%	9.01%	LSTM (+0.36%)
Vacation Period	210	11.53%	6.64%	Persistence (+4.89%)

Figure: Comparative analysis of regime segment results



SHAP

Interpretability



SHAP Interpretability

XAI Methodology

2

1

Results



SHAP Interpretability

XAI
Methodology

2

1

Results

- Bridging blind predictions with interpretable feature contributions allows grid dispatchers to see why an anomaly or surge is predicted and act upon it
- Gradient Explainer guarantees local accuracy and mathematical consistency
- $(200,48,18) \rightarrow (200,18)$



SHAP Interpretability

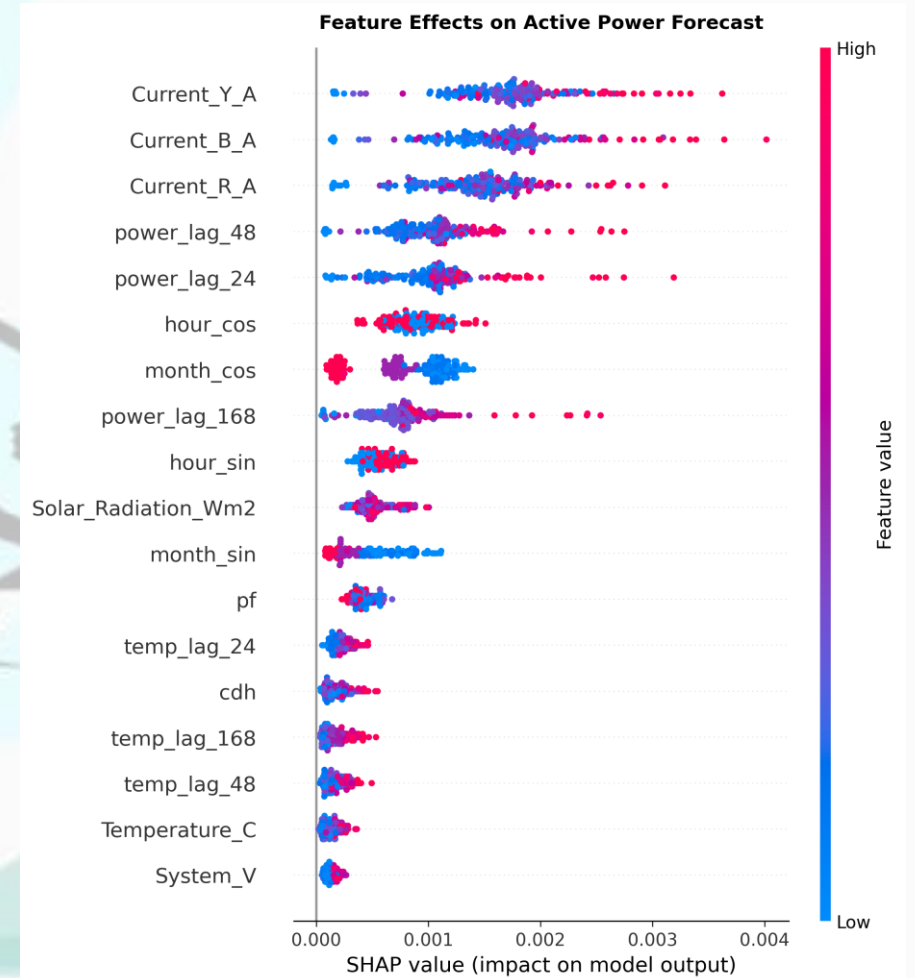
XAI Methodology

2

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Results

- Recurring load patterns from previous days and weeks in addition to current measurements, dominate as expected
- Voltage and power factor provide relatively little additional predictive information once current and load history are available





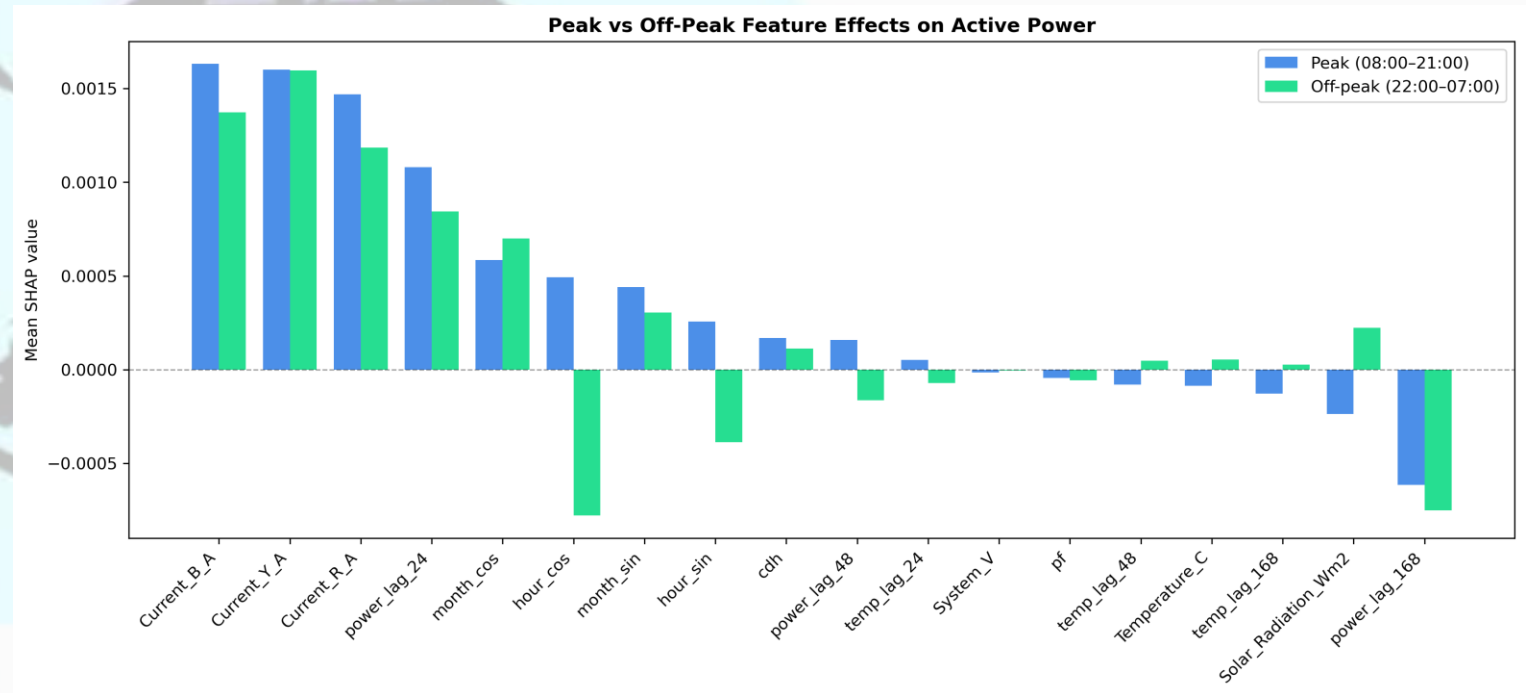
SHAP Interpretability

XAI Methodology

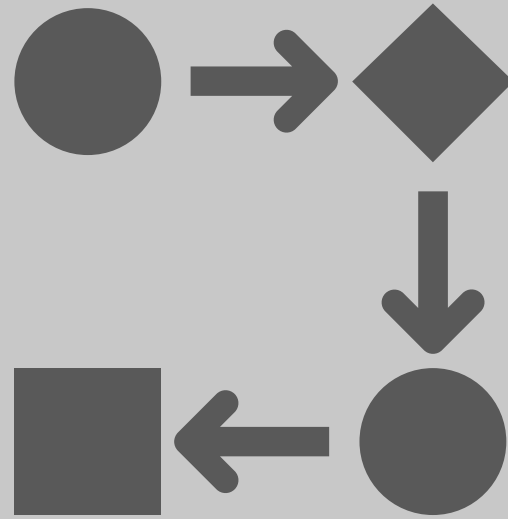
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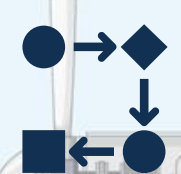
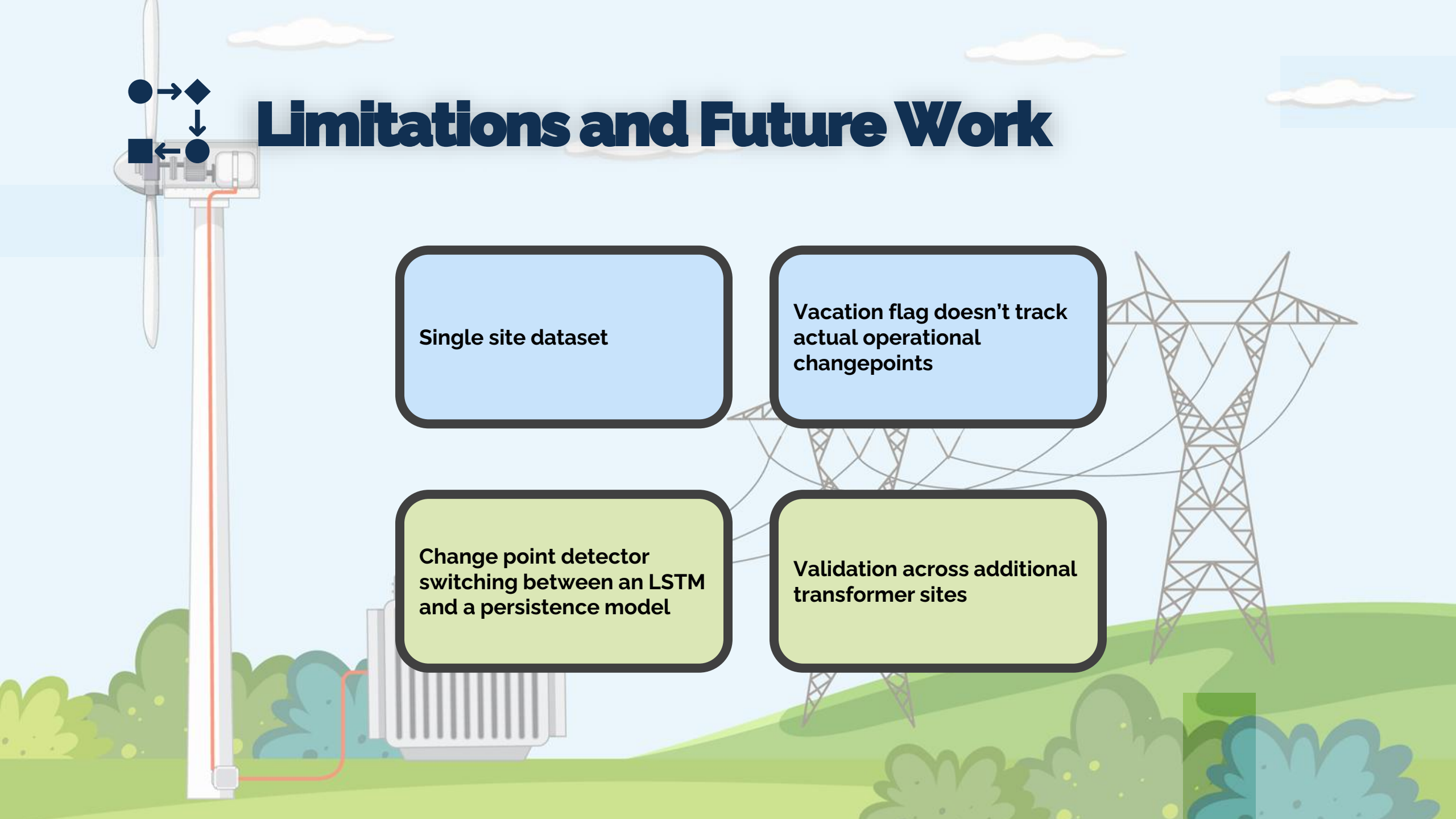
Results



- time-dependent changes in demand behaviour are being captured
- Solar radiation has moderate importance while temperature is weak



Limitations and Future Work



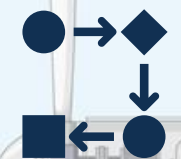
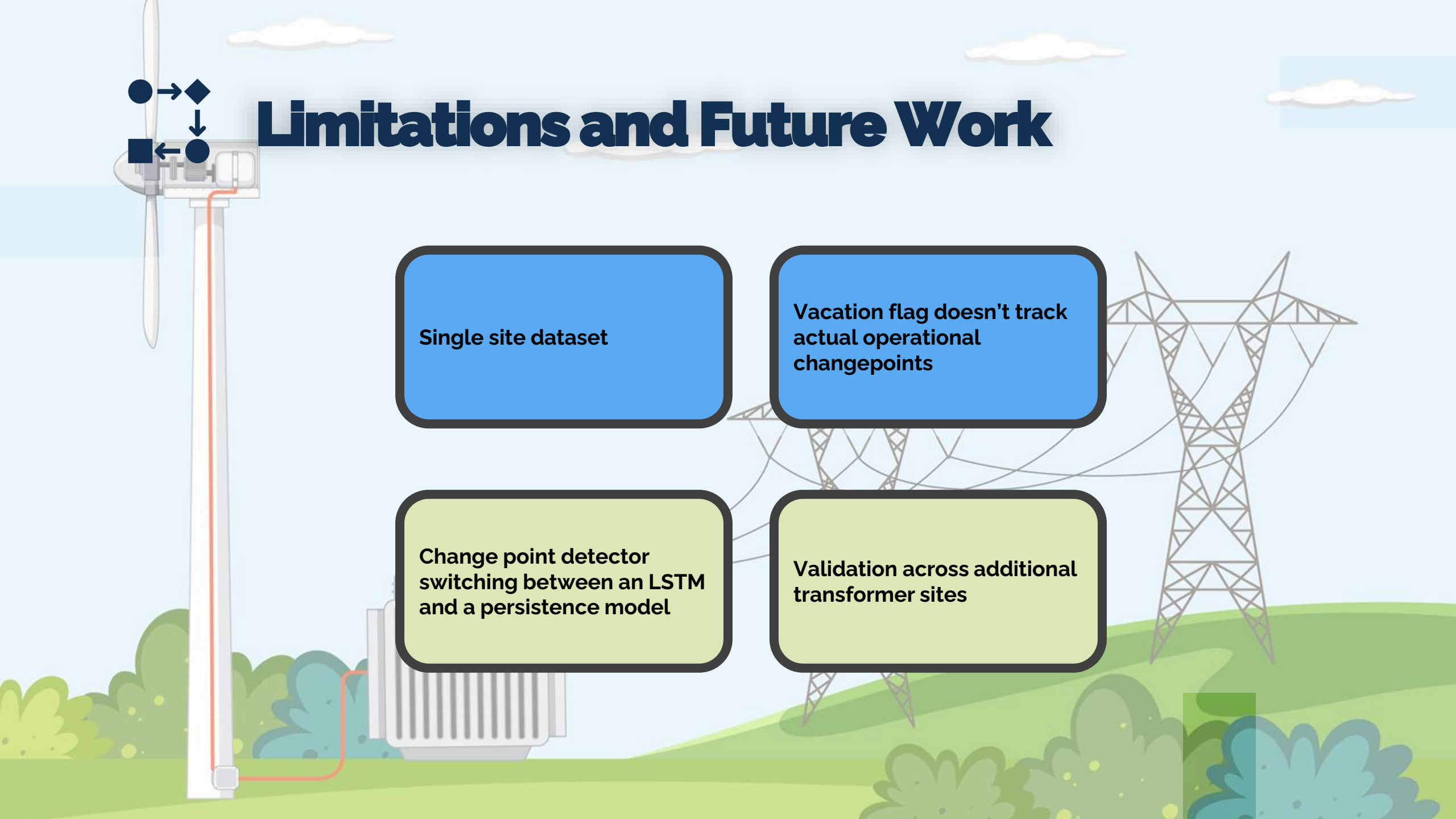
Limitations and Future Work

Single site dataset

Vacation flag doesn't track actual operational changepoints

Change point detector switching between an LSTM and a persistence model

Validation across additional transformer sites



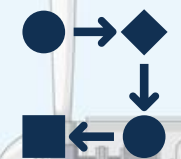
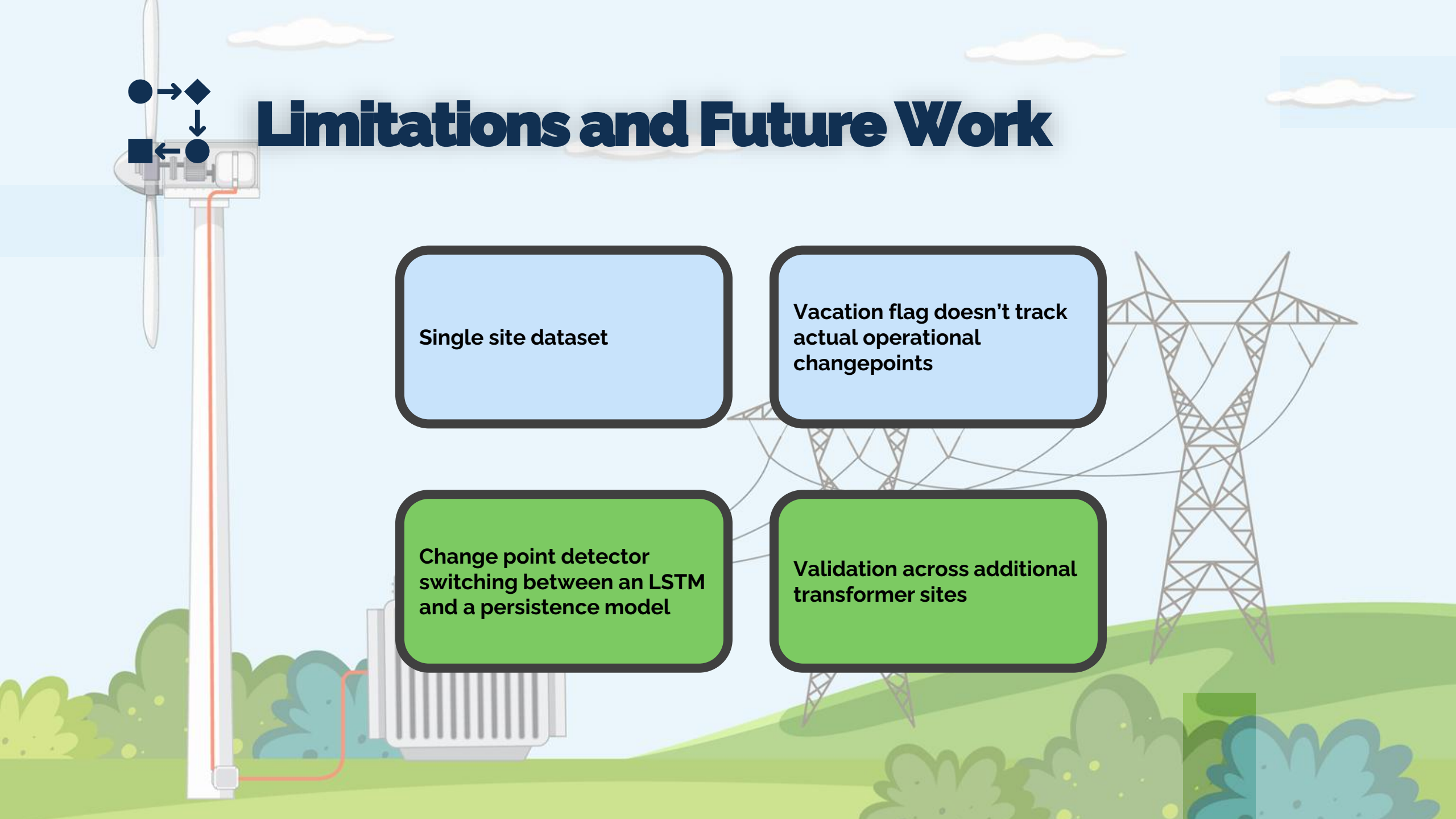
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Conclusion



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ACCURACY & BENCHMARKING

Overall Accuracy of 8.87% with a steady-state forecasting advantage over the naive persistence baseline (7.65% vs. 9.66% MAPE).



INTERPRETABILITY

Model transparency helps grid operators optimize generation scheduling and capacity planning.

Model transparency helps understand time of day plays a crucial role in demand change from peak/off-peak hours,



SCIENTIFIC CONTRIBUTION

Addressing the regional research gap for South Asian smart grid operations under tropical climate profiles.

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